



# St. Francis Institute of Technology

(Engineering College)

**An Autonomous Institute, Affiliated to University of Mumbai**

NAAC A+ Accredited | CMPN, EXTC, INFT NBA Accredited | ISO 9001:2015 Certified

## **Department of Information Technology**

**Class: SE-ITA/ITB Semester: IV; A.Y. 2024-2025**

**Subject: Microprocessor Lab (ITL 403)**

### **Experiment – 8: Compute factorial of a positive number**

#### **1. Aim:**

Write an ALP to compute the factorial of a positive number.

#### **2. Requirements**

DOSBox (an x86 emulator with DOS), Turbo Assembler, Turbo Debugger

#### **3. Pre-Experiment Exercise**

##### **Algorithm:**

- A.** Initialize the data segment.
- B.** Display message “enter the number to calculate factorial:\$” and use INT 21h commands to take a single digit input from the user.
- C.** Call a procedure to calculate the factorial of the number.
- D.** Store the result in data memory.

#### **4. Laboratory Exercise:**

##### **Procedure:**

- A.** Open DOSbox and go to TASM.
- B.** Open a new document using the command - edit <filename>.asm
- C.** Write the Program and save the changes to the same file.
- D.** Assemble the program using the command - tasm <filename.asm>
- E.** If any errors are displayed, then change the code in <filename>
- F.** If no errors are displayed, execute the command - tlink <filename>.obj to create the executable file.
- G.** Next execute the command - td <filename>
- H.** Try to RUN the program step by step and view the changes in the registers, flags, memory, etc.

## 5. Post Experiment Exercise:

### A. Results/Calculations/Observations:

Attach appropriate screenshots of the input from the user and output stored in memory along with the ALP.

### B. Questions:

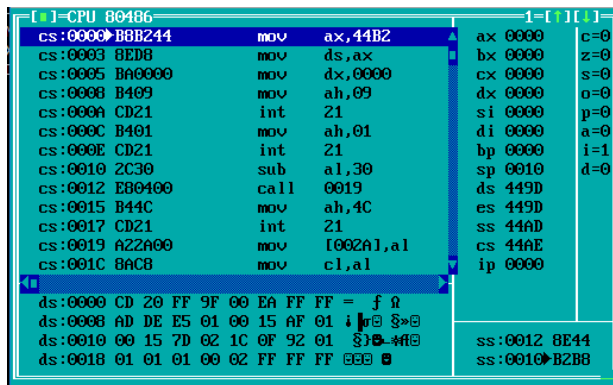
- Write an ALP in TASM to calculate LCM of two 16-bit numbers. Attach appropriate screenshots.
- Write an ALP in TASM to calculate GCD of two 16-bit numbers. Attach appropriate screenshots.

### C. Conclusion:

Write the conclusion/comments based on the experiment performed and the output obtained.

### D. References:

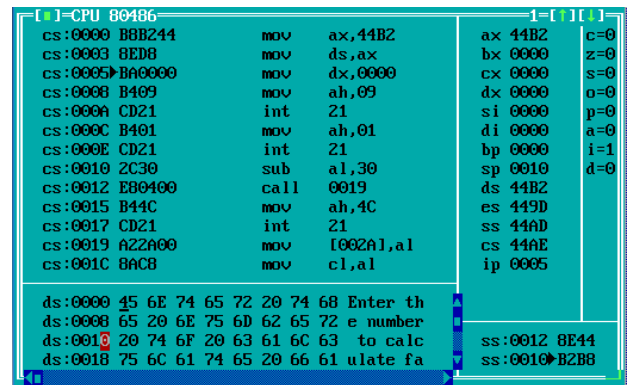
Mention two book references and two web references.



Initial state of registers and memory:

| Register | Value |
|----------|-------|
| ax       | 0000  |
| bx       | 0000  |
| cx       | 0000  |
| dx       | 0000  |
| si       | 0000  |
| di       | 0000  |
| bp       | 0000  |
| sp       | 0010  |
| ds       | 449D  |
| es       | 449D  |
| ss       | 44AD  |
| cs       | 44AE  |
| ip       | 0000  |

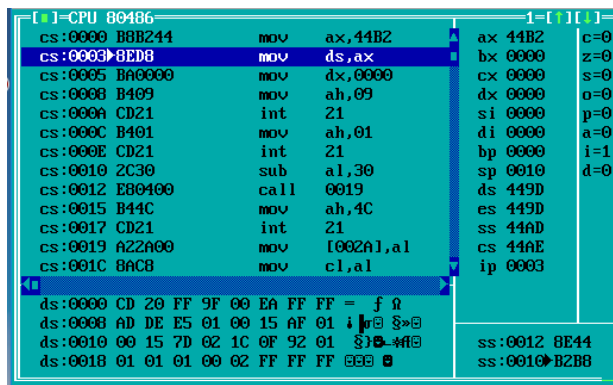
Observations: The debugger starts



Initial state of registers and memory:

| Register | Value |
|----------|-------|
| ax       | 44B2  |
| bx       | 0000  |
| cx       | 0000  |
| dx       | 0000  |
| si       | 0000  |
| di       | 0000  |
| bp       | 0000  |
| sp       | 0010  |
| ds       | 44B2  |
| es       | 449D  |
| ss       | 44AD  |
| cs       | 44AE  |
| ip       | 0005  |

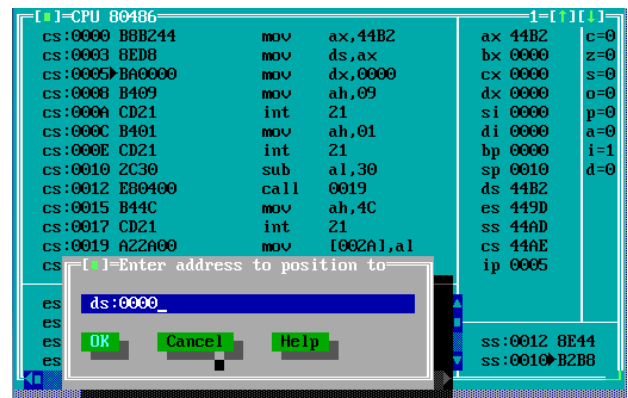
Observations: DX register set to 0000h



Initial state of registers and memory:

| Register | Value |
|----------|-------|
| ax       | 44B2  |
| bx       | 0000  |
| cx       | 0000  |
| dx       | 0000  |
| si       | 0000  |
| di       | 0000  |
| bp       | 0000  |
| sp       | 0010  |
| ds       | 44B2  |
| es       | 449D  |
| ss       | 44AD  |
| cs       | 44AE  |
| ip       | 0003  |

Observations: Value in AX is transferred to DS, setting up the data segment



Initial state of registers and memory:

| Register | Value |
|----------|-------|
| ax       | 44B2  |
| bx       | 0000  |
| cx       | 0000  |
| dx       | 0000  |
| si       | 0000  |
| di       | 0000  |
| bp       | 0000  |
| sp       | 0010  |
| ds       | 44B2  |
| es       | 449D  |
| ss       | 44AD  |
| cs       | 44AE  |
| ip       | 0005  |

```

[CPU 80486]
cs:0000 B8B244    mov ax,44B2    ax 44B2    c=0
cs:0003 8ED8      mov ds,ax      bx 0000    z=0
cs:0005 BA0000    mov dx,0000    cx 0000    s=0
cs:0008 B409      mov ah,09      dx 0000    o=0
cs:000A CD21      int 21      si 0000    p=0
cs:000C B401      mov ah,01      di 0000    a=0
cs:000E CD21      int 21      bp 0000    i=1
cs:0010 2C30      sub al,30      sp 0010    d=0
cs:0012 E80400    call 0019      ds 44B2
cs:0015 B44C      mov ah,4C      es 449D
cs:0017 CD21      int 21      ss 44AD
cs:0019 A22A00    mov [002A],al  cs 44AE
cs:001C 8AC8      mov cl,al      ip 0008

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0012 BE44
ss:0010 B2B8

```

Observations: AH register is loaded with 09h to prepare DOS interrupt that displays string

```

[CPU 80486]
cs:0000 B8B244    mov ax,44B2    ax 01B2    c=0
cs:0003 8ED8      mov ds,ax      bx 0000    z=0
cs:0005 BA0000    mov dx,0000    cx 0000    s=0
cs:0008 B409      mov ah,09      dx 0000    o=0
cs:000A CD21      int 21      si 0000    p=0
cs:000C B401      mov ah,01      di 0000    a=0
cs:000E CD21      int 21      bp 0000    i=1
cs:0010 2C30      sub al,30      sp 0010    d=0
cs:0012 E80400    call 0019      ds 44B2
cs:0015 B44C      mov ah,4C      es 449D
cs:0017 CD21      int 21      ss 44AD
cs:0019 A22A00    mov [002A],al  cs 44AE
cs:001C 8AC8      mov cl,al      ip 000E

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0012 BE44
ss:0010 B2B8

```

Observations:

```

C:\TASM\td Exp8.exe
Turbo Debugger Version 2.51 Copyright (c) 1988,91 Borland International
Enter the number to calculate factorial:

```

Observations: System waits for user to enter number

```

[CPU 80486]
cs:0000 B8B244    mov ax,44B2    ax 09B2    c=0
cs:0003 8ED8      mov ds,ax      bx 0000    z=0
cs:0005 BA0000    mov dx,0000    cx 0000    s=0
cs:0008 B409      mov ah,09      dx 0000    o=0
cs:000A CD21      int 21      si 0000    p=0
cs:000C B401      mov ah,01      di 0000    a=0
cs:000E CD21      int 21      bp 0000    i=1
cs:0010 2C30      sub al,30      sp 0010    d=0
cs:0012 E80400    call 0019      ds 44B2
cs:0015 B44C      mov ah,4C      es 449D
cs:0017 CD21      int 21      ss 44AD
cs:0019 A22A00    mov [002A],al  cs 44AE
cs:001C 8AC8      mov cl,al      ip 000A

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0012 BE44
ss:0010 B2B8

```

Observations: DOS interrupt 21h executes

```

[CPU 80486]
cs:0000 B8B244    mov ax,44B2    ax 0136    c=0
cs:0003 8ED8      mov ds,ax      bx 0000    z=0
cs:0005 BA0000    mov dx,0000    cx 0000    s=0
cs:0008 B409      mov ah,09      dx 0000    o=0
cs:000A CD21      int 21      si 0000    p=0
cs:000C B401      mov ah,01      di 0000    a=0
cs:000E CD21      int 21      bp 0000    i=1
cs:0010 2C30      sub al,30      sp 0010    d=0
cs:0012 E80400    call 0019      ds 44B2
cs:0015 B44C      mov ah,4C      es 449D
cs:0017 CD21      int 21      ss 44AD
cs:0019 A22A00    mov [002A],al  cs 44AE
cs:001C 8AC8      mov cl,al      ip 0010

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0012 BE44
ss:0010 B2B8

```

Observations: Entered number's ASCII value stored in AL

```

[CPU 80486]
cs:0000 B8B244    mov ax,44B2    ax 09B2    c=0
cs:0003 8ED8      mov ds,ax      bx 0000    z=0
cs:0005 BA0000    mov dx,0000    cx 0000    s=0
cs:0008 B409      mov ah,09      dx 0000    o=0
cs:000A CD21      int 21      si 0000    p=0
cs:000C B401      mov ah,01      di 0000    a=0
cs:000E CD21      int 21      bp 0000    i=1
cs:0010 2C30      sub al,30      sp 0010    d=0
cs:0012 E80400    call 0019      ds 44B2
cs:0015 B44C      mov ah,4C      es 449D
cs:0017 CD21      int 21      ss 44AD
cs:0019 A22A00    mov [002A],al  cs 44AE
cs:001C 8AC8      mov cl,al      ip 000C

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0012 BE44
ss:0010 B2B8

```

Observations: AH register is set to 01h to prepare for reading a single character input from user

```

[CPU 80486]
cs:0000 B8B244    mov ax,44B2    ax 0106    c=0
cs:0003 8ED8      mov ds,ax      bx 0000    z=0
cs:0005 BA0000    mov dx,0000    cx 0000    s=0
cs:0008 B409      mov ah,09      dx 0000    o=0
cs:000A CD21      int 21      si 0000    p=1
cs:000C B401      mov ah,01      di 0000    a=0
cs:000E CD21      int 21      bp 0000    i=1
cs:0010 2C30      sub al,30      sp 0010    d=0
cs:0012 E80400    call 0019      ds 44B2
cs:0015 B44C      mov ah,4C      es 449D
cs:0017 CD21      int 21      ss 44AD
cs:0019 A22A00    mov [002A],al  cs 44AE
cs:001C 8AC8      mov cl,al      ip 0012

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0012 BE44
ss:0010 B2B8

```

Observations: Procedure called at memory location 0019h to compute factorial

```

[.] CPU 80486 ds:002A = 00 1-[] []
cs:0000 B8B244 mov ax,44B2 ax 0106 c=0
cs:0003 8ED8 mov ds,ax bx 0000 z=0
cs:0005 BA0000 mov dx,0000 cx 0000 s=0
cs:0008 B409 mov ah,09 dx 0000 o=0
cs:000A CD21 int 21 si 0000 p=1
cs:000C B401 mov ah,01 di 0000 a=0
cs:000E CD21 int 21 bp 0000 i=1
cs:0010 2C30 sub al,30 sp 000E d=0
cs:0012 E80400 call 0019 ds 44B2
cs:0015 B44C mov ah,4C es 449D
cs:0017 CD21 int 21 ss 44AD
cs:0019 A22A00 mov [002A],al cs 44AE
cs:001C 8AC8 mov cl,al ip 0019

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa
ss:0010 B2B8
ss:000E 0015

```

Observations: Numeric value of AL stored in memory at 002A for later use

```

[.] CPU 80486 1-[] []
cs:001C 8AC8 mov cl,al ax 0106 c=0
cs:001E B500 mov ch,00 bx 0000 z=0
cs:0020 B80100 mov ax,0001 cx 0006 s=0
cs:0023 BA0000 mov dx,0000 dx 0000 o=0
cs:0026 F7E1 mul cx si 0000 p=1
cs:0028 E2FC loop 0026 di 0000 a=0
cs:002A A32B00 mov [002B],ax bp 0000 i=1
cs:002D 89162D00 mov [002D],dx sp 000E d=0
cs:0031 C3 ret ds 44B2
cs:0032 0000 add [bx+si],al es 449D
cs:0034 0000 add [bx+si],al ss 44AD
cs:0036 0000 add [bx+si],al cs 44AE
cs:0038 0000 add [bx+si],al ip 0020

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa
ss:0010 B2B8
ss:000E 0015

```

Observations: The AX register is initialized to 0001h, ensuring that the factorial computation starts with 1.

```

[.] CPU 80486 1-[] []
cs:001C 8AC8 mov cl,al ax 0106 c=0
cs:001E B500 mov ch,00 bx 0000 z=0
cs:0020 B80100 mov ax,0001 cx 0006 s=0
cs:0023 BA0000 mov dx,0000 dx 0000 o=0
cs:0026 F7E1 mul cx si 0000 p=1
cs:0028 E2FC loop 0026 di 0000 a=0
cs:002A A32B00 mov [002B],ax bp 0000 i=1
cs:002D 89162D00 mov [002D],dx sp 000E d=0
cs:0031 C3 ret ds 44B2
cs:0032 0000 add [bx+si],al es 449D
cs:0034 0000 add [bx+si],al ss 44AD
cs:0036 0000 add [bx+si],al cs 44AE
cs:0038 0000 add [bx+si],al ip 001C

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa
ss:0010 B2B8
ss:000E 0015

```

Observations: The value in AL is copied to CL, setting CL = 05h for loop counting.

```

[.] CPU 80486 1-[] []
cs:001C 8AC8 mov cl,al ax 0001 c=0
cs:001E B500 mov ch,00 bx 0000 z=0
cs:0020 B80100 mov ax,0001 cx 0006 s=0
cs:0023 BA0000 mov dx,0000 dx 0000 o=0
cs:0026 F7E1 mul cx si 0000 p=1
cs:0028 E2FC loop 0026 di 0000 a=0
cs:002A A32B00 mov [002B],ax bp 0000 i=1
cs:002D 89162D00 mov [002D],dx sp 000E d=0
cs:0031 C3 ret ds 44B2
cs:0032 0000 add [bx+si],al es 449D
cs:0034 0000 add [bx+si],al ss 44AD
cs:0036 0000 add [bx+si],al cs 44AE
cs:0038 0000 add [bx+si],al ip 0023

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa
ss:0010 B2B8
ss:000E 0015

```

Observations: The DX register is cleared to ensure no residual values interfere with multiplication.

```

[.] CPU 80486 1-[] []
cs:001C 8AC8 mov cl,al ax 0106 c=0
cs:001E B500 mov ch,00 bx 0000 z=0
cs:0020 B80100 mov ax,0001 cx 0006 s=0
cs:0023 BA0000 mov dx,0000 dx 0000 o=0
cs:0026 F7E1 mul cx si 0000 p=1
cs:0028 E2FC loop 0026 di 0000 a=0
cs:002A A32B00 mov [002B],ax bp 0000 i=1
cs:002D 89162D00 mov [002D],dx sp 000E d=0
cs:0031 C3 ret ds 44B2
cs:0032 0000 add [bx+si],al es 449D
cs:0034 0000 add [bx+si],al ss 44AD
cs:0036 0000 add [bx+si],al cs 44AE
cs:0038 0000 add [bx+si],al ip 001E

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa
ss:0010 B2B8
ss:000E 0015

```

Observations: The CH register is cleared, ensuring that CX is correctly set as 0006h.

```

[.] CPU 80486 1-[] []
cs:001C 8AC8 mov cl,al ax 0001 c=0
cs:001E B500 mov ch,00 bx 0000 z=0
cs:0020 B80100 mov ax,0001 cx 0006 s=0
cs:0023 BA0000 mov dx,0000 dx 0000 o=0
cs:0026 F7E1 mul cx si 0000 p=1
cs:0028 E2FC loop 0026 di 0000 a=0
cs:002A A32B00 mov [002B],ax bp 0000 i=1
cs:002D 89162D00 mov [002D],dx sp 000E d=0
cs:0031 C3 ret ds 44B2
cs:0032 0000 add [bx+si],al es 449D
cs:0034 0000 add [bx+si],al ss 44AD
cs:0036 0000 add [bx+si],al cs 44AE
cs:0038 0000 add [bx+si],al ip 0026

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa
ss:0010 B2B8
ss:000E 0015

```

Observations: The value in CX is multiplied with AX. Since AX initially holds 0001h, the result is:  $AX = 1 \times 6 = 6$ ,  $DX = 0000h$

```

[.] CPU 80486
cs:001C 8AC8      mov     cl,al
cs:001E B500      mov     ch,00
cs:0020 B80100    mov     ax,0001
cs:0023 BA0000    mov     dx,0000
cs:0026 F7E1      mul     cx
cs:0028 E2FC      loop    0026 ↑
cs:002A A32B00    mov     [002B],ax
cs:002D 89162D00  mov     [002D],dx
cs:0031 C3        ret
cs:0032 0000      add     [bx+si],al
cs:0034 0000      add     [bx+si],al
cs:0036 0000      add     [bx+si],al
cs:0038 0000      add     [bx+si],al

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0010 B2B8
ss:000E 0015

```

Observations: CX is decremented to 0005h, and execution jumps back to 0026h since CX is not zero.

```

[.] CPU 80486
cs:001C 8AC8      mov     cl,al
cs:001E B500      mov     ch,00
cs:0020 B80100    mov     ax,0001
cs:0023 BA0000    mov     dx,0000
cs:0026 F7E1      mul     cx
cs:0028 E2FC      loop    0026 ↑
cs:002A A32B00    mov     [002B],ax
cs:002D 89162D00  mov     [002D],dx
cs:0031 C3        ret
cs:0032 0000      add     [bx+si],al
cs:0034 0000      add     [bx+si],al
cs:0036 0000      add     [bx+si],al
cs:0038 0000      add     [bx+si],al

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0010 B2B8
ss:000E 0015

```

Observations: : The new value of CX (0004h) is multiplied with AX:  $AX:AX = 30 \times 4 = 120$ ,  $DX = 0000h$ .

```

[.] CPU 80486
cs:001C 8AC8      mov     cl,al
cs:001E B500      mov     ch,00
cs:0020 B80100    mov     ax,0001
cs:0023 BA0000    mov     dx,0000
cs:0026 F7E1      mul     cx
cs:0028 E2FC      loop    0026 ↑
cs:002A A32B00    mov     [002B],ax
cs:002D 89162D00  mov     [002D],dx
cs:0031 C3        ret
cs:0032 0000      add     [bx+si],al
cs:0034 0000      add     [bx+si],al
cs:0036 0000      add     [bx+si],al
cs:0038 0000      add     [bx+si],al

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0010 B2B8
ss:000E 0015

```

Observations: The new value of CX is multiplied with AX:  $AX = 6 \times 5 = 30$ ,  $DX = 0000h$ .

```

[.] CPU 80486
cs:001C 8AC8      mov     cl,al
cs:001E B500      mov     ch,00
cs:0020 B80100    mov     ax,0001
cs:0023 BA0000    mov     dx,0000
cs:0026 F7E1      mul     cx
cs:0028 E2FC      loop    0026 ↑
cs:002A A32B00    mov     [002B],ax
cs:002D 89162D00  mov     [002D],dx
cs:0031 C3        ret
cs:0032 0000      add     [bx+si],al
cs:0034 0000      add     [bx+si],al
cs:0036 0000      add     [bx+si],al
cs:0038 0000      add     [bx+si],al

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0010 B2B8
ss:000E 0015

```

Observations: CX is decremented to 0003h, and execution jumps back to 0026h.

```

[.] CPU 80486
cs:001C 8AC8      mov     cl,al
cs:001E B500      mov     ch,00
cs:0020 B80100    mov     ax,0001
cs:0023 BA0000    mov     dx,0000
cs:0026 F7E1      mul     cx
cs:0028 E2FC      loop    0026 ↑
cs:002A A32B00    mov     [002B],ax
cs:002D 89162D00  mov     [002D],dx
cs:0031 C3        ret
cs:0032 0000      add     [bx+si],al
cs:0034 0000      add     [bx+si],al
cs:0036 0000      add     [bx+si],al
cs:0038 0000      add     [bx+si],al

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0010 B2B8
ss:000E 0015

```

Observations: : CX is decremented to 0004h, and execution jumps back to 0026h.

```

[.] CPU 80486
cs:001C 8AC8      mov     cl,al
cs:001E B500      mov     ch,00
cs:0020 B80100    mov     ax,0001
cs:0023 BA0000    mov     dx,0000
cs:0026 F7E1      mul     cx
cs:0028 E2FC      loop    0026 ↑
cs:002A A32B00    mov     [002B],ax
cs:002D 89162D00  mov     [002D],dx
cs:0031 C3        ret
cs:0032 0000      add     [bx+si],al
cs:0034 0000      add     [bx+si],al
cs:0036 0000      add     [bx+si],al
cs:0038 0000      add     [bx+si],al

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0010 B2B8
ss:000E 0015

```

Observations: The new value of CX (0003h) is multiplied with AX:  $AX = 120 \times 3 = 360$ ,  $DX = 0000h$ .

```

[CPU 80486]
cs:001C 8AC8      mov     cl,al
cs:001E B500      mov     ch,00
cs:0020 B80100    mov     ax,0001
cs:0023 BA0000    mov     dx,0000
cs:0026 F7E1      mul     cx
cs:0028 E2FC      loop    0026 ↑
cs:002A A32B00    mov     [002B],ax
cs:002D 89162D00  mov     [002D],dx
cs:0031 C3        ret
cs:0032 0000      add     [bx+si],al
cs:0034 0000      add     [bx+si],al
cs:0036 0000      add     [bx+si],al
cs:0038 0000      add     [bx+si],al

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0010 B2B8
ss:000E 0015

```

Observations: : CX is decremented to 0002h, and execution jumps back to 0026h.

```

[CPU 80486]
cs:001C 8AC8      mov     cl,al
cs:001E B500      mov     ch,00
cs:0020 B80100    mov     ax,0001
cs:0023 BA0000    mov     dx,0000
cs:0026 F7E1      mul     cx
cs:0028 E2FC      loop    0026
cs:002A A32B00    mov     [002B],ax
cs:002D 89162D00  mov     [002D],dx
cs:0031 C3        ret
cs:0032 0000      add     [bx+si],al
cs:0034 0000      add     [bx+si],al
cs:0036 0000      add     [bx+si],al
cs:0038 0000      add     [bx+si],al

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0010 B2B8
ss:000E 0015

```

Observations: The new value of CX (0001h) is multiplied with AX:  $AX = 720 \times 1 = 720$ ,  $DX = 0000h$ .

```

[CPU 80486]
cs:001C 8AC8      mov     cl,al
cs:001E B500      mov     ch,00
cs:0020 B80100    mov     ax,0001
cs:0023 BA0000    mov     dx,0000
cs:0026 F7E1      mul     cx
cs:0028 E2FC      loop    0026
cs:002A A32B00    mov     [002B],ax
cs:002D 89162D00  mov     [002D],dx
cs:0031 C3        ret
cs:0032 0000      add     [bx+si],al
cs:0034 0000      add     [bx+si],al
cs:0036 0000      add     [bx+si],al
cs:0038 0000      add     [bx+si],al

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0010 B2B8
ss:000E 0015

```

Observations: The new value of CX (0002h) is multiplied with AX:  $AX = 360 \times 2 = 720$ ,  $DX = 0000h$ .

```

[CPU 80486]
cs:001C 8AC8      mov     cl,al
cs:001E B500      mov     ch,00
cs:0020 B80100    mov     ax,0001
cs:0023 BA0000    mov     dx,0000
cs:0026 F7E1      mul     cx
cs:0028 E2FC      loop    0026
cs:002A A32B00    mov     [002B],ax
cs:002D 89162D00  mov     [002D],dx
cs:0031 C3        ret
cs:0032 0000      add     [bx+si],al
cs:0034 0000      add     [bx+si],al
cs:0036 0000      add     [bx+si],al
cs:0038 0000      add     [bx+si],al

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0010 B2B8
ss:000E 0015

```

Observations: : CX is decremented to 0000h, so the loop terminates, and execution proceeds to the next instruction.

```

[CPU 80486]
cs:001C 8AC8      mov     cl,al
cs:001E B500      mov     ch,00
cs:0020 B80100    mov     ax,0001
cs:0023 BA0000    mov     dx,0000
cs:0026 F7E1      mul     cx
cs:0028 E2FC      loop    0026
cs:002A A32B00    mov     [002B],ax
cs:002D 89162D00  mov     [002D],dx
cs:0031 C3        ret
cs:0032 0000      add     [bx+si],al
cs:0034 0000      add     [bx+si],al
cs:0036 0000      add     [bx+si],al
cs:0038 0000      add     [bx+si],al

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0010 B2B8
ss:000E 0015

```

Observations: CX is decremented to 0001h, and execution jumps back to 0026h.

```

[CPU 80486]
cs:001C 8AC8      mov     cl,al
cs:001E B500      mov     ch,00
cs:0020 B80100    mov     ax,0001
cs:0023 BA0000    mov     dx,0000
cs:0026 F7E1      mul     cx
cs:0028 E2FC      loop    0026
cs:002A A32B00    mov     [002B],ax
cs:002D 89162D00  mov     [002D],dx
cs:0031 C3        ret
cs:0032 0000      add     [bx+si],al
cs:0034 0000      add     [bx+si],al
cs:0036 0000      add     [bx+si],al
cs:0038 0000      add     [bx+si],al

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0010 B2B8
ss:000E 0015

```

Observations: The computed factorial result ( $AX = 02D0h$ ) is stored in memory location 002Bh.



```

[CPU 80486] ds:002D = 0000
cs:001C 8AC8      mov     cl,al
cs:001E B500      mov     ch,00
cs:0020 B80100    mov     ax,0001
cs:0023 BA0000    mov     dx,0000
cs:0026 F7E1      mul     cx
cs:0028 E2FC      loop    0026
cs:002A A32B00    mov     [002B],ax
cs:002D 89162D00  mov     [002D],dx
cs:0031 C3        ret
cs:0032 0000      add     [bx+sil,al]
cs:0034 0000      add     [bx+sil,al]
cs:0036 0000      add     [bx+sil,al]
cs:0038 0000      add     [bx+sil,al]

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0010 B2B8
ss:000E 0015

```

Observations: The DX register is stored at memory location 002Dh.

```

[CPU 80486]
cs:001C 8AC8      mov     cl,al
cs:001E B500      mov     ch,00
cs:0020 B80100    mov     ax,0001
cs:0023 BA0000    mov     dx,0000
cs:0026 F7E1      mul     cx
cs:0028 E2FC      loop    0026
cs:002A A32B00    mov     [002B],ax
cs:002D 89162D00  mov     [002D],dx
cs:0031 C3        ret
cs:0032 0000      add     [bx+sil,al]
cs:0034 0000      add     [bx+sil,al]
cs:0036 0000      add     [bx+sil,al]
cs:0038 0000      add     [bx+sil,al]

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0010 B2B8
ss:000E 0015

```

Observations: The procedure returns control to the caller, completing the factorial computation process. The result 720 can now be used for further operations or display.

```

[CPU 80486]
cs:0015 B44C      mov     ah,4C
cs:0017 CD21      int     21
cs:0019 A22A00    mov     [002A],al
cs:001C 8AC8      mov     cl,al
cs:001E B500      mov     ch,00
cs:0020 B80100    mov     ax,0001
cs:0023 BA0000    mov     dx,0000
cs:0026 F7E1      mul     cx
cs:0028 E2FC      loop    0026
cs:002A A32B00    mov     [002B],ax
cs:002D 89162D00  mov     [002D],dx
cs:0031 C3        ret
cs:0032 0000      add     [bx+sil,al]

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0012 8E44
ss:0010 B2B8

```

Observations: The AH register is set to 4Ch, which is the DOS interrupt function for program termination.

```

[CPU 80486]
cs:0015 B44C      mov     ah,4C
cs:0017 CD21      int     21
cs:0019 A22A00    mov     [002A],al
cs:001C 8AC8      mov     cl,al
cs:001E B500      mov     ch,00
cs:0020 B80100    mov     ax,0001
cs:0023 BA0000    mov     dx,0000
cs:0026 F7E1      mul     cx
cs:0028 E2FC      loop    0026
cs:002A A32B00    mov     [002B],ax
cs:002D 89162D00  mov     [002D],dx
cs:0031 C3        ret
cs:0032 0000      add     [bx+sil,al]

ds:0000 45 6E 74 65 72 20 74 68 Enter th
ds:0008 65 20 6E 75 6D 62 65 72 e number
ds:0010 20 74 6F 20 63 61 6C 63 to calc
ds:0018 75 6C 61 74 65 20 66 61 ulate fa

ss:0012 8E44
ss:0010 B2B8

```

Observations: : The program execution is terminated, and control is returned to the DOS environment. The system halts further execution, completing the factorial calculation process.

```

[CPU 80486]
44AE:0015 B44C      mov     ah,4C
44AE:0017 CD21      int     21
44AE:0019 A22A00    mov     [002A],al
44AE:001C 8AC8      mov     cl,al
44AE:001E B500      mov     ch,00
44AE:0020 B80100    mov     ax,0001
44AE:0023 BA0000    mov     dx,0000
44AE:0026 F7E1      mul     cx
44AE:0028 E2FC      loop    0026
44AE:002A A32B00    mov     [002B],ax
44AE:002D 89162D00  mov     [002D],dx
44AE:0031 C3        ret
44AE:0032 0000      add     [bx+sil,al]

44B2:0000 45 6E 74 65 72 20 74 68 Enter th
44B2:0008 65 20 6E 75 6D 62 65 72 e number
44B2:0010 20 74 6F 20 63 61 6C 63 to calc
44B2:0018 75 6C 61 74 65 20 66 61 ulate fa

44AD:0012 8E44
44AD:0010 B2B8

```

Observations: Code finished

DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: TD

File View Run Breakpoints Data Options Window Help

1-CPU 80486

```

cs:0000 B8B041      mov     ax,41B0      ax 001E      c=0
cs:0003 8ED8        mov     ds,ax        bx 000F      z=0
cs:0005 A10000       mov     ax,[0000]    cx 0000      s=0
cs:0008 B81E0200     mov     bx,[0002]    dx 0000      o=0
cs:000C 0BC8        mov     cx,ax        si 0000      p=0
cs:000E B8B3        mov     dx,bx        di 0000      a=0
cs:0010 3BC3        cmp     ax,bx        bp 0000      i=1
cs:0012 720A        jb      001E         sp 0400      d=0
cs:0014 7304        jnb     001A         ds 44B0
cs:0016 03C1        add     ax,cx        es 449D
cs:0018 EBF6        jmp     0010         ss 44B1
cs:001A 03DA        add     bx,dx        cs 44AD
cs:001C EBF2        jmp     0010         ip 000C

```

```

ds:0000 1E 00 0F 00 00 00 00 00 00 00 ▲ *
ds:0008 00 00 00 00 00 00 00 00 00 00
ds:0010 00 00 00 00 00 00 00 00 00 00
ds:0018 00 00 00 00 00 00 00 00 00 00

```

```

ss:0402 0000
ss:0400 0000

```







```

DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: TD
File View Run Breakpoints Data Options Window Help
[ ] CPU 80486
cs:0010 3BC3      cmp     ax,bx      ax 001E c=1
cs:0012 720A      jb      001E      bx 002D z=0
cs:0014 7304      jnb     001A      cx 001E s=1
cs:0016 03C1      add     ax,cx      dx 000F o=0
cs:0018 EBF6      jmp     0010      si 0000 p=0
cs:001A 03DA      add     bx,dx      di 0000 a=0
cs:001C EBF2      jmp     0010      bp 0000 i=1
cs:001E A30400    mov     [0004],ax  sp 0400 d=0
cs:0021 CD21      int     21        ds 44B0
cs:0023 0000      add     [bx+si],al es 449D
cs:0025 0000      add     [bx+si],al ss 44B1
cs:0027 0000      add     [bx+si],al cs 44AD
cs:0029 0000      add     [bx+si],al ip 0012

ds:0000 1E 00 0F 00 00 00 00  *
ds:0008 00 00 00 00 00 00 00
ds:0010 00 00 00 00 00 00 00
ds:0018 00 00 00 00 00 00 00
ss:0402 0000
ss:0400 0000

```

```

mov ax,@data
mov ds,ax
mov ax,num1
mov bx,num2
back:
cmp ax,bx
je result
jnc ahead
sub bx,ax
jmp back
ahead:
sub ax,bx
jmp back
result:
mov gcd,ax
int 21h
end

```

```

DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: TD
File View Run Breakpoints Data Options Window Help
[ ] CPU 80486
cs:0010 3BC3      cmp     ax,bx      ax 001E c=1
cs:0012 720A      jb      001E      bx 002D z=0
cs:0014 7304      jnb     001A      cx 001E s=1
cs:0016 03C1      add     ax,cx      dx 000F o=0
cs:0018 EBF6      jmp     0010      si 0000 p=0
cs:001A 03DA      add     bx,dx      di 0000 a=0
cs:001C EBF2      jmp     0010      bp 0000 i=1
cs:001E A30400    mov     [0004],ax  sp 0400 d=0
cs:0021 CD21      int     21        ds 44B0
cs:0023 0000      add     [bx+si],al es 449D
cs:0025 0000      add     [bx+si],al ss 44B1
cs:0027 0000      add     [bx+si],al cs 44AD
cs:0029 0000      add     [bx+si],al ip 001E

ds:0000 1E 00 0F 00 00 00 00  *
ds:0008 00 00 00 00 00 00 00
ds:0010 00 00 00 00 00 00 00
ds:0018 00 00 00 00 00 00 00
ss:0402 0000
ss:0400 0000

```

```

DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: TD
File View Run Breakpoints Data Options Window Help
[ ] CPU 80486
cs:0000 B8AF41    mov     ax,44AF    ax 0000 c=0
cs:0003 8ED8      mov     ds,ax      bx 0000 z=0
cs:0005 A10000    mov     ax,[0000]  cx 0000 s=0
cs:0008 8B1E0200  mov     bx,[0002]  dx 0000 o=0
cs:000C 3BC3      cmp     ax,bx      si 0000 p=0
cs:000E 740A      je      001A      di 0000 a=0
cs:0010 7304      jnb     0016      bp 0000 i=1
cs:0012 2BD8      sub     bx,ax      sp 0400 d=0
cs:0014 EBF6      jmp     000C      es 449D
cs:0016 2BC3      sub     ax,bx      ss 44B0
cs:0018 EBF2      jmp     000C      cs 44AD
cs:001A A30400    mov     [0004],ax  ip 0000
cs:001D CD21      int     21

ds:0000 CD 20 FF 9F 00 EA FF FF = f 0
ds:0008 AD DE E5 01 00 15 AF 01 i 0 0 0 0
ds:0010 00 15 7D 02 1C 0F 92 01 0 0 0 0
ds:0018 01 01 01 00 02 FF FF FF 0 0 0 0
ss:0402 0000
ss:0400 0000

```

```

DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: TD
File View Run Breakpoints Data Options Window Help
[ ] CPU 80486
cs:0010 3BC3      cmp     ax,bx      ax 001E c=1
cs:0012 720A      jb      001E      bx 002D z=0
cs:0014 7304      jnb     001A      cx 001E s=1
cs:0016 03C1      add     ax,cx      dx 000F o=0
cs:0018 EBF6      jmp     0010      si 0000 p=0
cs:001A 03DA      add     bx,dx      di 0000 a=0
cs:001C EBF2      jmp     0010      bp 0000 i=1
cs:001E A30400    mov     [0004],ax  sp 0400 d=0
cs:0021 CD21      int     21        ds 44B0
cs:0023 0000      add     [bx+si],al es 449D
cs:0025 0000      add     [bx+si],al ss 44B1
cs:0027 0000      add     [bx+si],al cs 44AD
cs:0029 0000      add     [bx+si],al ip 0021

ds:0000 1E 00 0F 00 1E 00 00  * *
ds:0008 00 00 00 00 00 00 00
ds:0010 00 00 00 00 00 00 00
ds:0018 00 00 00 00 00 00 00
ss:0402 0000
ss:0400 0000

```

```

DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: TD
File View Run Breakpoints Data Options Window Help
[ ] CPU 80486
cs:0000 B8AF41    mov     ax,44AF    ax 44AF c=0
cs:0003 8ED8      mov     ds,ax      bx 0000 z=0
cs:0005 A10000    mov     ax,[0000]  cx 0000 s=0
cs:0008 8B1E0200  mov     bx,[0002]  dx 0000 o=0
cs:000C 3BC3      cmp     ax,bx      si 0000 p=0
cs:000E 740A      je      001A      di 0000 a=0
cs:0010 7304      jnb     0016      bp 0000 i=1
cs:0012 2BD8      sub     bx,ax      sp 0400 d=0
cs:0014 EBF6      jmp     000C      es 449D
cs:0016 2BC3      sub     ax,bx      ss 44B0
cs:0018 EBF2      jmp     000C      cs 44AD
cs:001A A30400    mov     [0004],ax  ip 0003
cs:001D CD21      int     21

ds:0000 CD 20 FF 9F 00 EA FF FF = f 0
ds:0008 AD DE E5 01 00 15 AF 01 i 0 0 0 0
ds:0010 00 15 7D 02 1C 0F 92 01 0 0 0 0
ds:0018 01 01 01 00 02 FF FF FF 0 0 0 0
ss:0402 0000
ss:0400 0000

```

```

2)GCD
.model small
.stack
.data
num1 dW 0001Eh
num2 dW 000Fh
gcd dW 01 dup(?)
.code

```



DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: TD

| File View Run Breakpoints Data Options Window Help |          |               |             |
|----------------------------------------------------|----------|---------------|-------------|
| [CPU 80486]                                        |          |               |             |
| cs:0000                                            | B8AF44   | mov ax,44AF   | ax 000F c=0 |
| cs:0003                                            | 8ED8     | mov ds,ax     | bx 000F z=0 |
| cs:0005                                            | A10000   | mov ax,[0000] | cx 0000 s=0 |
| cs:0008                                            | BB1E0200 | mov bx,[0002] | dx 0000 o=0 |
| cs:000C                                            | 3BC3     | cmp ax,bx     | si 0000 p=1 |
| cs:000E                                            | 740A     | je 001A       | di 0000 a=1 |
| cs:0010                                            | 7304     | jnb 0016      | bp 0000 i=1 |
| cs:0012                                            | 2BD8     | sub bx,ax     | sp 0400 d=0 |
| cs:0014                                            | EBF6     | jmp 000C      | ds 44AF     |
| cs:0016                                            | 2BC3     | sub ax,bx     | es 449D     |
| cs:0018                                            | EBF2     | jmp 000C      | ss 44B0     |
| cs:001A                                            | A30400   | mov [0004],ax | cs 44AD     |
| cs:001D                                            | CD21     | int 21        | ip 0018     |
| ds:0000 1E 00 0F 00 00 00 00 ▲ *                   |          |               |             |
| ds:0008 00 00 00 00 00 00 00                       |          |               |             |
| ds:0010 00 00 00 00 00 00 00                       |          |               |             |
| ds:0018 00 00 00 00 00 00 00                       |          |               |             |
| ss:0402 0000                                       |          |               |             |
| ss:0400 0000                                       |          |               |             |

DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: TD

| File View Run Breakpoints Data Options Window Help |          |               |             |
|----------------------------------------------------|----------|---------------|-------------|
| [CPU 80486]                                        |          |               |             |
| cs:0000                                            | B8AF44   | mov ax,44AF   | ax 000F c=0 |
| cs:0003                                            | 8ED8     | mov ds,ax     | bx 000F z=1 |
| cs:0005                                            | A10000   | mov ax,[0000] | cx 0000 s=0 |
| cs:0008                                            | BB1E0200 | mov bx,[0002] | dx 0000 o=0 |
| cs:000C                                            | 3BC3     | cmp ax,bx     | si 0000 p=1 |
| cs:000E                                            | 740A     | je 001A       | di 0000 a=0 |
| cs:0010                                            | 7304     | jnb 0016      | bp 0000 i=1 |
| cs:0012                                            | 2BD8     | sub bx,ax     | sp 0400 d=0 |
| cs:0014                                            | EBF6     | jmp 000C      | ds 44AF     |
| cs:0016                                            | 2BC3     | sub ax,bx     | es 449D     |
| cs:0018                                            | EBF2     | jmp 000C      | ss 44B0     |
| cs:001A                                            | A30400   | mov [0004],ax | cs 44AD     |
| cs:001D                                            | CD21     | int 21        | ip 001A     |
| ds:0000 1E 00 0F 00 00 00 00 ▲ *                   |          |               |             |
| ds:0008 00 00 00 00 00 00 00                       |          |               |             |
| ds:0010 00 00 00 00 00 00 00                       |          |               |             |
| ds:0018 00 00 00 00 00 00 00                       |          |               |             |
| ss:0402 0000                                       |          |               |             |
| ss:0400 0000                                       |          |               |             |

DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: TD

| File View Run Breakpoints Data Options Window Help |          |               |             |
|----------------------------------------------------|----------|---------------|-------------|
| [CPU 80486]                                        |          |               |             |
| cs:0000                                            | B8AF44   | mov ax,44AF   | ax 000F c=0 |
| cs:0003                                            | 8ED8     | mov ds,ax     | bx 000F z=0 |
| cs:0005                                            | A10000   | mov ax,[0000] | cx 0000 s=0 |
| cs:0008                                            | BB1E0200 | mov bx,[0002] | dx 0000 o=0 |
| cs:000C                                            | 3BC3     | cmp ax,bx     | si 0000 p=1 |
| cs:000E                                            | 740A     | je 001A       | di 0000 a=1 |
| cs:0010                                            | 7304     | jnb 0016      | bp 0000 i=1 |
| cs:0012                                            | 2BD8     | sub bx,ax     | sp 0400 d=0 |
| cs:0014                                            | EBF6     | jmp 000C      | ds 44AF     |
| cs:0016                                            | 2BC3     | sub ax,bx     | es 449D     |
| cs:0018                                            | EBF2     | jmp 000C      | ss 44B0     |
| cs:001A                                            | A30400   | mov [0004],ax | cs 44AD     |
| cs:001D                                            | CD21     | int 21        | ip 000C     |
| ds:0000 1E 00 0F 00 00 00 00 ▲ *                   |          |               |             |
| ds:0008 00 00 00 00 00 00 00                       |          |               |             |
| ds:0010 00 00 00 00 00 00 00                       |          |               |             |
| ds:0018 00 00 00 00 00 00 00                       |          |               |             |
| ss:0402 0000                                       |          |               |             |
| ss:0400 0000                                       |          |               |             |

DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: TD

| File View Run Breakpoints Data Options Window Help |          |                |             |
|----------------------------------------------------|----------|----------------|-------------|
| [CPU 80486]                                        |          |                |             |
| cs:001D                                            | CD21     | int 21         | ax 000F c=0 |
| cs:001F                                            | 001E000F | add [0F00],b1  | bx 000F z=1 |
| cs:0023                                            | 000F     | add [bx],c1    | cx 0000 s=0 |
| cs:0025                                            | 0000     | add [bx+si],a1 | dx 0000 o=0 |
| cs:0027                                            | 0000     | add [bx+si],a1 | si 0000 p=1 |
| cs:0029                                            | 0000     | add [bx+si],a1 | di 0000 a=0 |
| cs:002B                                            | 0000     | add [bx+si],a1 | bp 0000 i=1 |
| cs:002D                                            | 0000     | add [bx+si],a1 | sp 0400 d=0 |
| cs:002F                                            | 0000     | add [bx+si],a1 | ds 44AF     |
| cs:0031                                            | 0000     | add [bx+si],a1 | es 449D     |
| cs:0033                                            | 0000     | add [bx+si],a1 | ss 44B0     |
| cs:0035                                            | 0000     | add [bx+si],a1 | cs 44AD     |
| cs:0037                                            | 0000     | add [bx+si],a1 | ip 001D     |
| ds:0000 1E 00 0F 00 0F 00 00 ▲ * *                 |          |                |             |
| ds:0008 00 00 00 00 00 00 00                       |          |                |             |
| ds:0010 00 00 00 00 00 00 00                       |          |                |             |
| ds:0018 00 00 00 00 00 00 00                       |          |                |             |
| ss:0402 0000                                       |          |                |             |
| ss:0400 0000                                       |          |                |             |

DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: TD

| File View Run Breakpoints Data Options Window Help |          |               |             |
|----------------------------------------------------|----------|---------------|-------------|
| [CPU 80486]                                        |          |               |             |
| cs:0000                                            | B8AF44   | mov ax,44AF   | ax 000F c=0 |
| cs:0003                                            | 8ED8     | mov ds,ax     | bx 000F z=1 |
| cs:0005                                            | A10000   | mov ax,[0000] | cx 0000 s=0 |
| cs:0008                                            | BB1E0200 | mov bx,[0002] | dx 0000 o=0 |
| cs:000C                                            | 3BC3     | cmp ax,bx     | si 0000 p=1 |
| cs:000E                                            | 740A     | je 001A       | di 0000 a=0 |
| cs:0010                                            | 7304     | jnb 0016      | bp 0000 i=1 |
| cs:0012                                            | 2BD8     | sub bx,ax     | sp 0400 d=0 |
| cs:0014                                            | EBF6     | jmp 000C      | ds 44AF     |
| cs:0016                                            | 2BC3     | sub ax,bx     | es 449D     |
| cs:0018                                            | EBF2     | jmp 000C      | ss 44B0     |
| cs:001A                                            | A30400   | mov [0004],ax | cs 44AD     |
| cs:001D                                            | CD21     | int 21        | ip 000E     |
| ds:0000 1E 00 0F 00 00 00 00 ▲ *                   |          |               |             |
| ds:0008 00 00 00 00 00 00 00                       |          |               |             |
| ds:0010 00 00 00 00 00 00 00                       |          |               |             |
| ds:0018 00 00 00 00 00 00 00                       |          |               |             |
| ss:0402 0000                                       |          |               |             |
| ss:0400 0000                                       |          |               |             |